

Problem 19 ✓ Tangent lines to an ellipse at the endpoints of a diam. are $\parallel$.

$$E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$P(x_0, y_0)$ one endpoint of diam.

$P'(-x_0, -y_0)$ other " — "

$$\Rightarrow m_P \left(\frac{x_0}{a^2}, \frac{y_0}{b^2} \right)$$

Tangent at P : $T_{(x_0, y_0)} E_{a,b}: \frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$

at P' : $T_{(-x_0, -y_0)} E_{a,b}: -\frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1 \quad | \cdot (-1)$

$$\Rightarrow \frac{xx_0}{a^2} + \frac{yy_0}{b^2} = -1 \Rightarrow m_{P'} \left(\frac{x_0}{a^2}, \frac{y_0}{b^2} \right)$$

$\Rightarrow m_P = m_{P'} \Rightarrow$ same normal vector $\Rightarrow$ parallel tangent lines.

Problem 20 ✓ Hyperbola: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, tangent lines of a given slope k
 $\Delta = 0$

$$\begin{cases} H_{a,b}: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \\ y = kx + m \end{cases}$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{(kx+m)^2}{b^2} = 1 \quad | \cdot a^2 b^2$$

$$\Rightarrow b^2 x^2 - a^2 (kx+m)^2 = a^2 b^2$$

$$\Rightarrow (b^2 - a^2 k^2) x^2 - 2kma^2 x - a^2(m^2 + b^2) = 0$$

Tangency $\Rightarrow$ double sol, eg. $\Rightarrow \Delta = 0$.

$$\Rightarrow \Delta = (2kma^2)^2 + 4(b^2 - a^2 k^2) \cdot a^2(m^2 + b^2) = 0$$

$$\Delta = \dots (a^2 k^2 - b^2 - m^2) = 0$$

$$\Rightarrow m^2 = a^2 k^2 - b^2$$

$$m = \pm \sqrt{a^2 k^2 - b^2}$$

$$y = kx + m = kx \pm \sqrt{a^2 k^2 - b^2}$$

Problem 21 ✓ Hyperbola: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, tangent line at given point (x_0, y_0)

$$\begin{cases} H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \end{cases}$$

$$l: \begin{cases} x = x_0 + t \cdot v_x \\ y = y_0 + t \cdot v_y \end{cases}$$

l is tangent to H when $|l \cap H| = 1$

$$\frac{(x_0 + t \cdot v_x)^2}{a^2} - \frac{(y_0 + t \cdot v_y)^2}{b^2} = 1$$

$$\Rightarrow \frac{x_0^2 + 2x_0 \cdot t \cdot v_x + t^2 \cdot v_x^2}{a^2} - \frac{y_0^2 + 2y_0 \cdot t \cdot v_y + t^2 \cdot v_y^2}{b^2} = 1$$

$$\Rightarrow t^2 \left(\frac{v_x^2}{a^2} - \frac{v_y^2}{b^2} \right) + 2t \left(\frac{x_0 \cdot v_x}{a^2} - \frac{y_0 \cdot v_y}{b^2} \right) + \underbrace{\frac{x_0^2}{a^2} - \frac{y_0^2}{b^2} - 1}_0 = 0$$

$$t=0 \text{ double sol } (\Rightarrow) \frac{x_0 \cdot v_x}{a^2} - \frac{y_0 \cdot v_y}{b^2} = 0$$

$$n \left(\frac{x_0}{a^2}, -\frac{y_0}{b^2} \right) \text{ normal vector for line } l.$$

$$l: \frac{x_0}{a^2} (x - x_0) + \frac{y_0}{b^2} (y - y_0) = 0$$

$$l: \frac{x x_0}{a^2} - \frac{x_0^2}{a^2} - \frac{y y_0}{b^2} + \frac{y_0^2}{b^2} = 0$$

$$l: \frac{x x_0}{a^2} - \frac{y y_0}{b^2} - \underbrace{\left(\frac{x_0^2}{a^2} - \frac{y_0^2}{b^2} \right)}_1 = 0$$

$$\Rightarrow J_{(x_0, y_0)} f_{a,b} : \frac{x x_0}{a^2} - \frac{y y_0}{b^2} = 1$$

Problem 22 Parabola $y^2 = 2px$ ($p > 0$), tangent line of given slope k has the equation $y = kx + \frac{p}{2k}$?

$$\begin{cases} P: y^2 = 2px \\ y = kx + m \end{cases}$$

$$\Rightarrow (kx + m)^2 = 2px$$

$$\Rightarrow k^2 x^2 + 2kmx + m^2 = 2px$$

$$\Rightarrow k^2 x^2 + (2km - 2p)x + m^2 = 0$$

$$\Delta = 0 \Rightarrow \dots \Rightarrow p - 2km = 0 \Rightarrow m = \frac{p}{2k}$$

$$y = kx + m = kx + \frac{p}{2k}$$

Problem 23 Parabola $y^2 = 2px$, tangent line at given point (x_0, y_0)

$$\begin{cases} P: y^2 = 2px \end{cases}$$

$$l: \begin{cases} x = x_0 + t \cdot v_x \\ y = y_0 + t \cdot v_y \end{cases}$$

$$\Rightarrow (y_0 + t \cdot v_y)^2 = 2p(x_0 + t \cdot v_x)$$

$$\Rightarrow t^2 \cdot v_y^2 + 2t \cdot y_0 \cdot v_y + y_0^2 - 2p v_x t - 2p \cdot x_0 = 0$$

$$\Rightarrow \cancel{v_y^2 \cdot t^2 + 2y_0 \cdot v_y \cdot t -}$$

$$\Rightarrow \underbrace{v_y^2 \cdot t^2 + (2y_0 \cdot v_y - 2p v_x)t + y_0^2 - 2p \cdot x_0}_0 = 0$$

$$2y_0 \cdot v_y - 2p v_x = 0 \Rightarrow y_0 \cdot v_y = p v_x \Rightarrow v_x - y_0 \cdot v_y = 0 \Rightarrow$$

$$\Rightarrow (1, -y_0) \text{ normal vector}$$

$$\Rightarrow l: x - x_0 - y_0(y - y_0) = 0 \Rightarrow l: p x - x_0 - y y_0 + y_0^2 = 0$$

Better with gradient

Exercise 24 ✓ $\Sigma_{a,b,c}: \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, tangent plane at point (x_0, y_0, z_0) .

$$\left\{ \begin{array}{l} \Sigma_{a,b,c}: \dots \\ l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} + t \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \end{array} \right. \quad \begin{array}{l} x = x_p + t \cdot v_x \\ y = y_p + t \cdot v_y \\ z = z_p + t \cdot v_z \end{array}$$

$$l \cap \Sigma \Leftrightarrow \frac{(x_p + t \cdot v_x)^2}{a^2} + \frac{(y_p + t \cdot v_y)^2}{b^2} + \frac{(z_p + t \cdot v_z)^2}{c^2} = 1$$

$$\Rightarrow t^2 \left(\frac{v_x^2}{a^2} + \frac{v_y^2}{b^2} + \frac{v_z^2}{c^2} \right) + 2t \left(\frac{x_p v_x}{a^2} + \frac{y_p v_y}{b^2} + \frac{z_p v_z}{c^2} \right) + \underbrace{\frac{x_p^2}{a^2} + \frac{y_p^2}{b^2} + \frac{z_p^2}{c^2}}_1 = 1$$

$$\Rightarrow \dots = 0$$

$$\Rightarrow 2 \left(\frac{x_p v_x}{a^2} + \frac{y_p v_y}{b^2} + \frac{z_p v_z}{c^2} \right) = 0$$

$\Rightarrow m \left(\frac{x_p}{a^2}, \frac{y_p}{b^2}, \frac{z_p}{c^2} \right)$ normal vector
(m is orthogonal to the plane containing lines l)

$$\Rightarrow \langle N, \begin{bmatrix} \frac{x_p}{a^2} \\ \frac{y_p}{b^2} \\ \frac{z_p}{c^2} \end{bmatrix} \rangle = 0$$

$\parallel$
 m

$$\nabla_p \Sigma_{a,b,c}: \frac{x_p}{a^2} (x - x_p) + \frac{y_p}{b^2} (y - y_p) + \frac{z_p}{c^2} (z - z_p) = 0$$

$$\nabla_p \Sigma_{a,b,c}: \begin{bmatrix} \frac{x_p}{a^2} \\ \frac{y_p}{b^2} \\ \frac{z_p}{c^2} \end{bmatrix} \cdot \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} - \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} \right) = 0$$

$$\nabla_p \Sigma_{a,b,c}: \frac{x x_p}{a^2} + \frac{y y_p}{b^2} + \frac{z z_p}{c^2} - \underbrace{\left(\frac{x_p^2}{a^2} + \frac{y_p^2}{b^2} + \frac{z_p^2}{c^2} \right)}_1 = 0$$

$$\nabla_p \Sigma_{a,b,c}: \frac{x x_p}{a^2} + \frac{y y_p}{b^2} + \frac{z z_p}{c^2} = 1.$$

Problem 25 ✓ Cone: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$; tangent plane at point (x_0, y_0, z_0)

$$\begin{cases} C_{a,b,c}: \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0 \\ l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \cdot \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} \end{cases} \uparrow$$

$$\frac{t^2 x_0^2}{a^2} + \frac{t^2 y_0^2}{b^2} - \frac{t^2 z_0^2}{c^2} = 0 \Leftrightarrow t^2 \left(\underbrace{\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - \frac{z_0^2}{c^2}}_0 \right) = 0.$$

$\Rightarrow l$ is a generator for the surface.

Proof with the gradient:

Define $F(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$

$C_{a,b,c}: F(x, y, z) = 0$

$\nabla F(x, y, z) = \left(\frac{2x}{a^2}, \frac{2y}{b^2}, -\frac{2z}{c^2} \right)$

at given point $P(x_0, y_0, z_0)$: $\nabla F(P) = \left(\frac{2x_0}{a^2}, \frac{2y_0}{b^2}, -\frac{2z_0}{c^2} \right) \Rightarrow$ normal vector

tangent plane: $\frac{2x_0}{a^2}(x-x_0) + \frac{2y_0}{b^2}(y-y_0) - \frac{2z_0}{c^2}(z-z_0) = 0 \quad | :2$

$\Rightarrow \frac{xx_0}{a^2} + \frac{yy_0}{b^2} - \frac{zz_0}{c^2} - \underbrace{\left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - \frac{z_0^2}{c^2} \right)}_0 = 0$

$\forall P \in C_{a,b,c}: \frac{xx_0}{a^2} + \frac{yy_0}{b^2} - \frac{zz_0}{c^2} = 0$

Problem 26 ✓ Cone: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$ - ruled surface?

$\forall x (x_0, y_0, z_0) \in C_{a,b,c}$

$l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \cdot \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$

$\left(\frac{tx_0}{a} \right)^2 + \left(\frac{ty_0}{b} \right)^2 - \left(\frac{tz_0}{c} \right)^2 = 0$

$\Leftrightarrow t^2 \cdot \left(\underbrace{\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - \frac{z_0^2}{c^2}}_0 \right) = 0 \Rightarrow l \text{ generator for } S. \text{ (surface)}$
(the whole line lies on the cone)

The line $l: \{ (tx_0, ty_0, tz_0) : t \in \mathbb{R} \}$ which passes through the given point and the origin is contained in $C_{a,b,c}$.

$\Rightarrow$ the set of all lines obtained in this way = generator of the cone.

If you can split it in two surfaces $\Rightarrow$ not a rule surface
(two sheets $\Rightarrow$ not a rule surface)

27) Come $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$

$\nabla \phi_{a,b,c}$

(*) $P \in \text{come}$, the generator through $P \subset$ tangent plane at P .

→ Generator through P : $(x,y,z) = t \cdot (x_0, y_0, z_0)$

→ $\nabla \phi_{a,b,c}$: $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} - \frac{zz_0}{c^2} = 0 \Rightarrow \dots + \left(\frac{y_0^2}{a^2} \dots \right) = 0$

$x = tx_0$

$y = ty_0$

$z = tz_0$

$\Rightarrow$ every point of the generator satisfies the tangent plane eq.

Problem 28 $\mathcal{H}^1$: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$, equations of one of the families of generators.

(hyperboloid)

$\frac{x^2}{a^2} - \frac{z^2}{c^2} = 1 - \frac{y^2}{b^2}$

$\Rightarrow \left(\frac{x}{a} - \frac{z}{c} \right) \left(\frac{x}{a} + \frac{z}{c} \right) = \left(1 - \frac{y}{b} \right) \left(1 + \frac{y}{b} \right)$

Divide one way:

$\frac{\frac{x}{a} - \frac{z}{c}}{1 - \frac{y}{b}} = \frac{1 + \frac{y}{b}}{\frac{x}{a} + \frac{z}{c}} = \alpha$

$l_\alpha: \begin{cases} \frac{x}{a} - \frac{z}{c} = \alpha \left(1 - \frac{y}{b} \right) \\ 1 + \frac{y}{b} = \alpha \left(\frac{x}{a} + \frac{z}{c} \right) \end{cases} \Rightarrow \text{line.}$

Problem 29 ✓ Hyperbolic paraboloid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - 2z = 0$, equation of the tangent plane at given point (x_0, y_0, z_0)

Proof (gradient)

① $(x_0, y_0, z_0) = \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - 2z_0 = 0$

$(\nabla \phi)(x_0, y_0, z_0) = \left(\frac{2x_0}{a^2}, \frac{2y_0}{b^2}, -2 \right)$ normal vector ($\nabla \perp$ level set).

$\frac{2x_0}{a^2} (x - x_0) + \frac{2y_0}{b^2} (y - y_0) - 2(z - z_0) = 0 \quad | :2$

$\Rightarrow \frac{xx_0}{a^2} + \frac{yy_0}{b^2} + z_0 - \left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} + z_0 \right) = 0$

$\Rightarrow \frac{xx_0}{a^2} + \frac{yy_0}{b^2} + z - \left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - 2z_0 \right) = 0$

$\Rightarrow \frac{xx_0}{a^2} + \frac{yy_0}{b^2} + z - \left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - 2z_0 \right) + 2z_0 = 0$

$\Rightarrow \nabla_{(x_0, y_0, z_0)} \phi_{a,b}^h: \frac{xx_0}{a^2} + \frac{yy_0}{b^2} + z - z_0 = 0.$